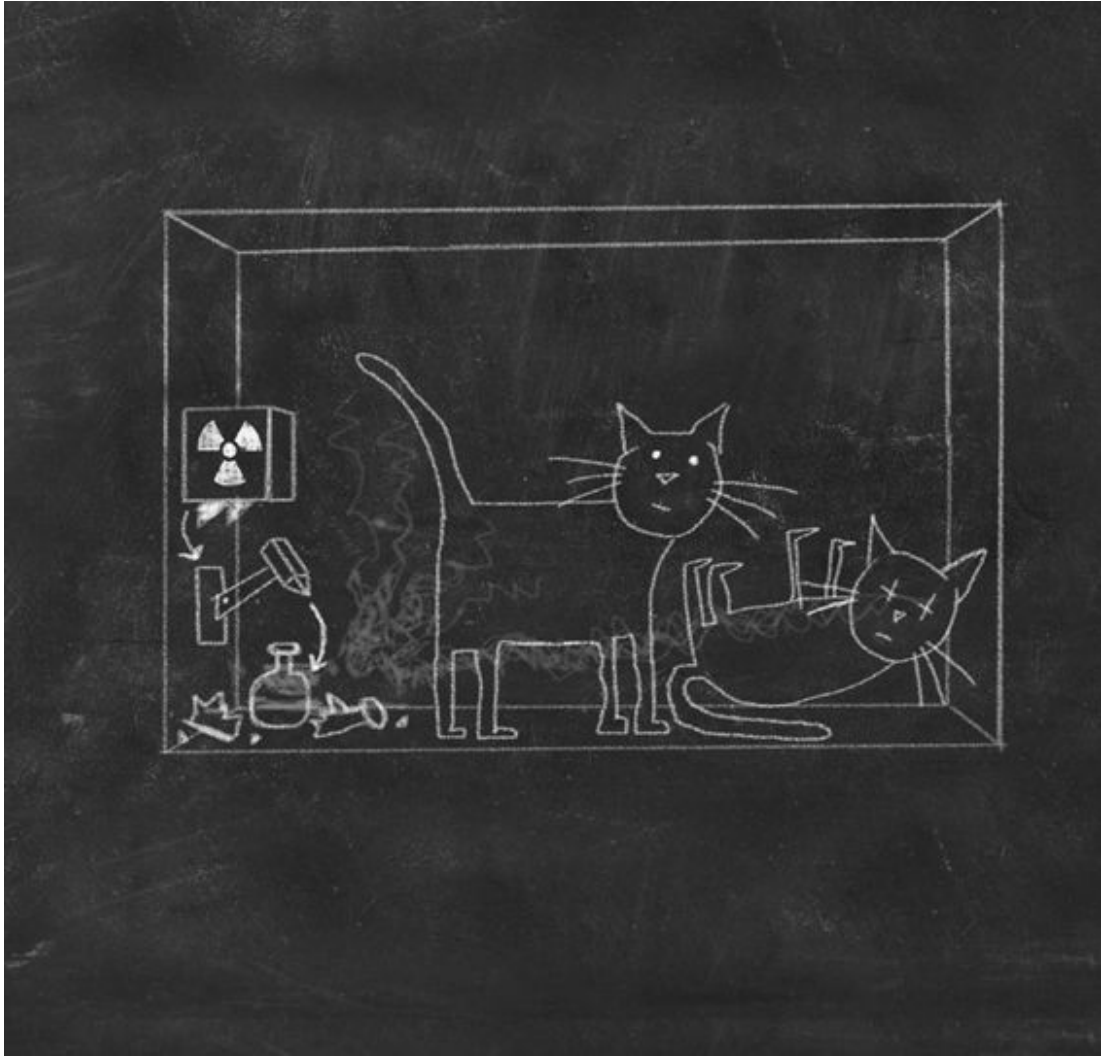


New philosophical questions

While Einstein's theories appeared to bring an end to a lot of metaphysical speculation, the physics that he pioneered also posed fresh questions. And rather than disproving old philosophical theories, it appeared to confirm or at least complement many. The definition of matter as energy, for example, has more than a superficial resemblance to the idealism of Schopenhauer or even Hindu and Buddhist philosophy. Many of the concepts of quantum mechanics, meanwhile, are hard for even physicists to understand, and seem almost mystical – such as Heisenberg's uncertainty principle and the 'observer effect' (the necessity of observation in 'fixing' the properties of a microscopic quantum system), which seem to defy common sense, but have an uncanny resemblance to Berkeley's philosophy. The Big Bang theory of the origin of the universe, meanwhile, also reopened the philosophical debate about the nature of reality and causality – and the possibility that there is more than one universe presents even more food for philosophical thought.



Quite apart from the scientific import of the famous ‘Schrödinger’s cat’ thought experiment, what are the philosophical implications of a cat being simultaneously alive and dead?